

Group Project Assignment

Course: Programming for Data Science (DS120)

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Date: April 13, 2026

Deadline: May 8

1. General Rules

- The project must be completed in groups of **up to 3 students**.
- Individual projects are **not allowed**.
- Within **1 week**, each group must:
 - Form a group
 - Send the instructor the **group member names**
 - Submit the **selected dataset**
- Work on the project can start **only after instructor approval**.

2. Project Objective

The objective of this project is to develop a **Python-based data analysis pipeline** that:

- Loads and preprocesses real-world data
- Performs structured data analysis
- Generates meaningful visualizations
- Follows a clean **object-oriented programming (OOP)** design

3. Project Components

3.1 Dataset Selection

- Select a real-world dataset (e.g., Kaggle, UCI, etc.)
- The dataset must be approved by the instructor
- The dataset should contain meaningful structure (categorical and numerical features preferred)

3.2 Data Loading and Preprocessing

- Load the dataset (any format)
- Handle missing values
- Remove duplicates
- Correct data types if needed
- Use **NumPy** and **Pandas**

3.3 Object-Oriented Design

- Implement at least **two classes**
- Example:
 - Dataset handling class
 - Analysis/visualization class
- Code must be modular, reusable, and well-structured

3.4 Data Analysis

- Implement at least **three analysis techniques**, such as:
 - Grouping and aggregation
 - Statistical summaries
 - Correlation analysis

3.5 Visualization

- Include at least **three different types of visualizations**
- Examples:
 - Distribution plots
 - Categorical plots
 - Relationship plots
- Use:
 - **Matplotlib** (required)
 - Seaborn (optional)

3.6 Code Quality and Documentation

- Follow **PEP8** standards
- Use docstrings for all classes and methods
- Organize code into modules (not a single script)

4. Deliverables

4.1 Code Submission

- Submit a zipped project folder
- Include all necessary modules, scripts, and data (if applicable)
- Submit via **Moodle**

4.2 Report

- A **1–2 page** written report including:
 - Objective of the analysis
 - Dataset description
 - Key insights
 - Overview of the code structure
- The report must be submitted via **Moodle Turnitin assignment**

5. Intermediate Attestation

- After **3 weeks**
- Worth **20%** of the final grade
- Students must demonstrate:
 - Data preprocessing progress
 - Initial analysis
 - Project structure and OOP design

6. Grading Breakdown

- Intermediate Attestation — 20%
- Data Preprocessing — 15%
- OOP Design and Code Structure — 15%
- Data Analysis — 15%
- Visualization — 15%

- Code Quality (PEP8, Documentation) — 10%
- Final Report — 10%

Total: 100 points

Important Note

All submitted work must be original. Any form of plagiarism or unauthorized collaboration will be penalized. Submissions will be checked using Turnitin.